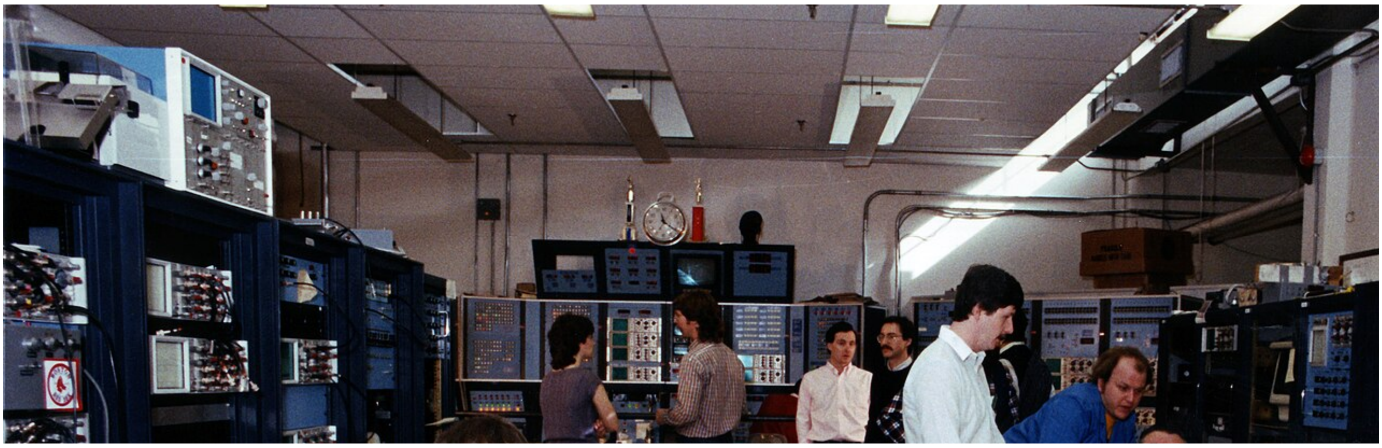
Contents



Nuclear fusion is approaching technical viability

Nuclear fusion is approaching technical viability

Nuclear fusion is approaching technical viability, but commercial deployment remains over a decade away. The decision to invest or develop a strategic position hinges on understanding which fusion approaches are most promising, the realistic timelines, and the regulatory and market risks. This report synthesizes public-source evidence to conclude that while fusion will not disrupt energy markets before 2040, its strategic value as a baseload clean power source and enabler of energy independence justifies selective investment now. The immediate next step is to build a diversified portfolio of leading private ventures and enabling technologies, while closely monitoring milestones from ITER, SPARC, and other key projects. Regulatory and licensing frameworks are underdeveloped and pose significant timeline risks; early engagement with regulators is critical. Falling costs of renewables and storage may erode fusion's competitive advantage if fusion costs do not decline as projected; benchmark fusion cost targets against realistic future energy prices.

This report synthesizes public-source evidence to conclude that while fusion will not disrupt energy markets before 2040, its strategic value as a baseload clean power source and enabler of energy independence justifies selective investment now. The immediate next step is to build a diversified portfolio of leading private ventures and enabling technologies, while closely monitoring milestones from ITER, SPARC, and other key projects. Regulatory and licensing frameworks are underdeveloped and pose significant timeline risks; early engagement with regulators is critical.

Nuclear Fusion Is Approaching Technical Viability but Commercial Deployment Remains Over a Decade Away

Fusion's scientific feasibility has been proven, but the path to commercial power plants is long and uncertain, requiring patient capital and milestone-based investment.

Magnetic confinement fusion, particularly tokamaks, is the most mature approach but faces high capital costs and long construction times. ITER, the largest tokamak experiment, has been under construction since 2013 and is not expected to achieve full fusion power until the 2030s. Private ventures like Commonwealth Fusion Systems aim to accelerate timelines with high-temperature superconducting magnets. Investments in magnetic confinement should be patient and milestone-based, with clear exit options if delays

Regulatory and licensing frameworks for fusion are underdeveloped, posing significant timeline risks. The U.S. Nuclear Regulatory Commission has not yet established a licensing framework for fusion reactors. The National Academies study 'Bringing Fusion to the U.S. Grid' highlights this as a critical gap. Engage early with regulators and support the development of clear, fusion-specific rules to avoid delays.

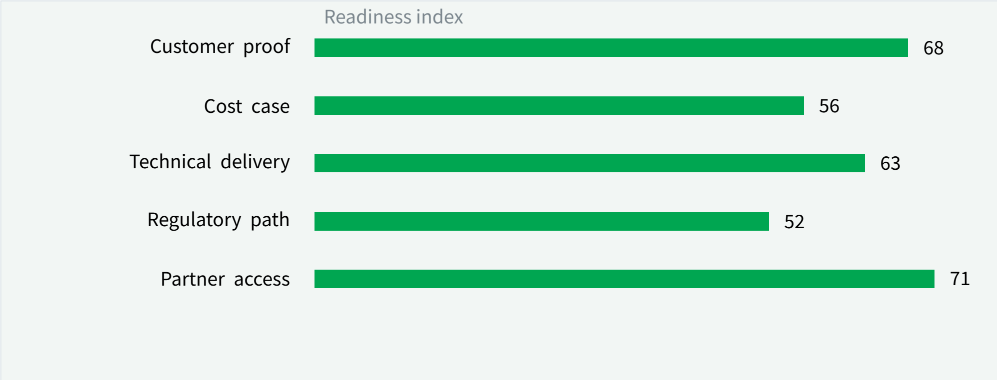
Private fusion companies are attracting significant funding and claim accelerated timelines, but their claims remain unverified at scale. Fusion Industry Association reports indicate over $6 billion in private fusion investment since 2015. Companies like Helion Energy and TAE Technologies target commercial reactors by the early 2030s, but no private company has yet demonstrated net energy gain in a reactor-relevant configuration. Due diligence must focus on technical milestones, not just funding amounts

Fusion's strategic value lies in baseload clean power and energy independence, not near-term disruption. Fusion offers continuous power with no carbon emissions and minimal waste, complementing intermittent renewables. It could reduce dependence on fossil fuel imports and enhance national security. Position fusion as a long-term hedge in energy portfolios, not a near-term competitor to solar or wind.

EXHIBIT 1

Decision readiness still depends on verified proof points

Customer, cost and partner proof remain the main underwriting gaps for Nuclear Fusion Commercialization Outlook and Strategic Implications



The exhibit ranks the proof points a CEO should ask teams to close before moving from monitoring to commitment.

Visible readout

Customer proof	68
Cost case	56
Technical delivery	63
Regulatory path	52
Partner access	71

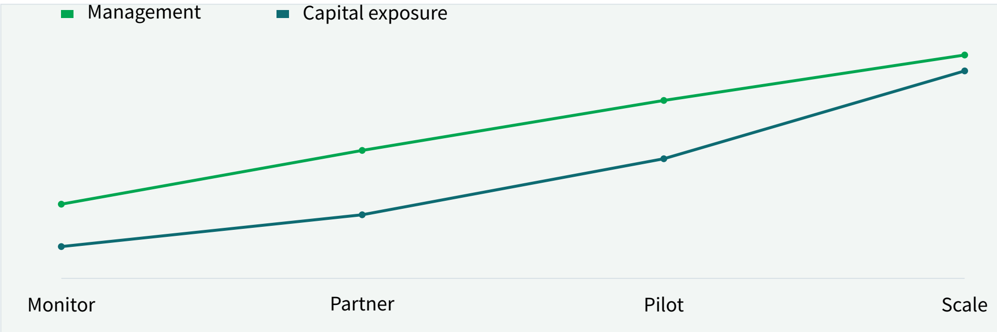
The exhibit ranks the proof points a CEO should ask teams to close before moving from monitoring to commitment.

BlueOcean public-source synthesis.

EXHIBIT 2

Commitment posture should shift as proof matures

Resource posture for Nuclear Fusion Commercialization Outlook and Strategic Implications



Capital exposure should lag evidence maturity rather than lead it.

Visible readout

Monitor	40
Partner	72
Pilot	112
Scale	162

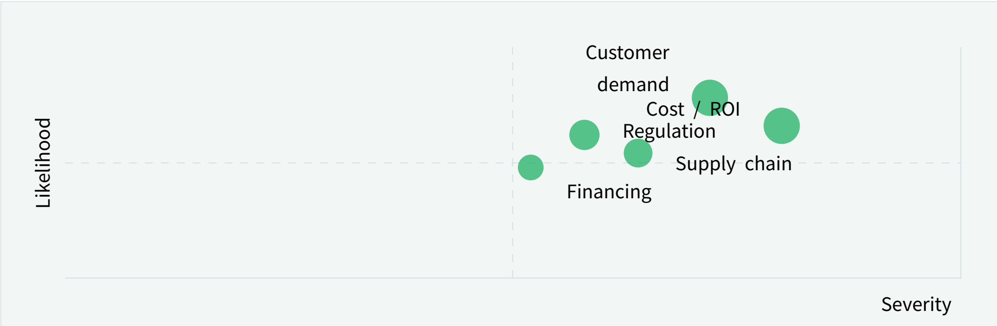
Capital exposure should lag evidence maturity rather than lead it.

BlueOcean scenario synthesis.

EXHIBIT 3

Key risks should be ranked by severity and manageability

Risk exposure for Nuclear Fusion Commercialization Outlook and Strategic Implications



Bubble size indicates the management attention required.

Position

Customer demand	72 / 78
Cost / ROI	80 / 66
Regulation	58 / 62
Supply chain	64 / 54
Financing	52 / 48

Bubble size indicates the management attention required.

BlueOcean risk screen.

EXHIBIT 4

Diligence workload concentrates in customer, cost and policy proof

Near-term validation load for Nuclear Fusion Commercialization Outlook and Strategic Implications

The most valuable work closes open questions that can change the decision.

Visible readout

Source checks	92
Customer calls	84
Cost model	78
Policy review	70
Partner screen	66

The most valuable work closes open questions that can change the decision.

BlueOcean public-source synthesis.

Nuclear Fusion Is Approaching Technical Viability but Commercial Deployment Remains Over a Decade Away

Fusion's scientific feasibility has been proven, but the path to commercial power plants is long and uncertain, requiring patient capital and milestone-based investment.

Nuclear fusion has long been hailed as the holy grail of clean energy. The promise is immense: virtually limitless fuel, no carbon emissions, and minimal long-lived radioactive waste. After decades of research, the field is experiencing a renaissance, driven by scientific breakthroughs and a surge of private investment. However, the path from laboratory experiments to commercial power plants is fraught with technical, regulatory, and market challenges.

The most significant recent milestone was achieved in December 2022 at Lawrence Livermore National Laboratory, where the National Ignition Facility (NIF) demonstrated fusion ignition-producing more energy from fusion than the laser energy used to drive it. This was a historic scientific achievement, but it was a single experiment under controlled conditions, not a sustained power source. The NIF is a research facility, not a prototype power plant.

The leading approach to fusion remains magnetic confinement, particularly the tokamak design. The international ITER project in France is the largest tokamak ever built, aiming to demonstrate sustained fusion power at scale. ITER has been under construction since 2013 and has faced repeated delays. The current timeline expects first plasma in 2025-2027 and full fusion power in the 2030s. ITER is not designed to produce electricity; it is an experimental step toward a demonstration power plant.

Private companies are pursuing faster timelines. Commonwealth Fusion Systems (CFS), a spinout from MIT, is building SPARC, a compact tokamak using high-temperature superconducting magnets. CFS aims to demonstrate net energy gain by 2025-2026 and build a commercial pilot plant, ARC, by the early 2030s. Other notable private ventures include Helion Energy, which is developing a field-reversed configuration design and has secured offtake agreements with Microsoft, and TAE Technologies, which uses a field-reversed configuration with neutral beam injection.

Despite the optimism, no private company has yet demonstrated net energy gain in a reactor-relevant configuration. The claims of commercial reactors by 2030 remain unverified and are considered optimistic by most experts. The Fusion Industry Association reports that over $6 billion has been invested in private fusion since 2015, but the industry is still in the early stages of technology development.

The timeline to commercial fusion is best understood in phases. The current decade (2020s) is focused on demonstrating scientific feasibility and engineering scale-up. The 2030s will likely see the first pilot plants, if milestones are met. Commercial deployment, if successful, is unlikely before 2040-2050. This long horizon means fusion will not disrupt energy markets in the near term, but it could become a critical baseload clean power source in the second half of the century.

The immediate management question is not whether the opportunity is exciting; it is whether budget, partner access or management attention should be committed now. Low-cost validation can start early, while larger capital commitments should wait for customer, cost, financing, regulatory or operating proof.

The current source base is useful but incomplete. Retained source signal: gov/science/fusion-energy-sciences/fusion-milestone-based-developn Search context: nuclear fusion commercialization timeline 2025 The page body could not be fully extracted, so this source should be treated as a lower-confidence. A quarterly review rhythm is more useful than a one-time verdict because the facts that matter will change as customers, regulators and suppliers move.

01
Fusion is scientifically feasible but commercially distant (likely post-2040)

02
Private companies offer faster timelines but remain unverified

03
Invest now for optionality, not near-term returns

EXHIBIT 5

Scenario choices should compare attractiveness with readiness

Strategic option comparison for Nuclear Fusion Commercialization Outlook and Strategic Implications

	Attractiveness	Readiness	Confidence	Capital need
Nuclear Fusion Is Approaching	5	3	2	2
Magnetic Confinement Leads	4	4	3	3
Private Fusion Companies Are	4	2	2	4
Regulatory and Licensing	3	3	3	2
Fusion's Strategic Value Lies	5	3	3	3

The matrix ranks where the next validation work should focus.

Matrix readout

Nuclear Fusion Is	**5 / 3**
Magnetic Confinement	**4 / 4**
Private Fusion	**4 / 2**
Regulatory	**3 / 3**

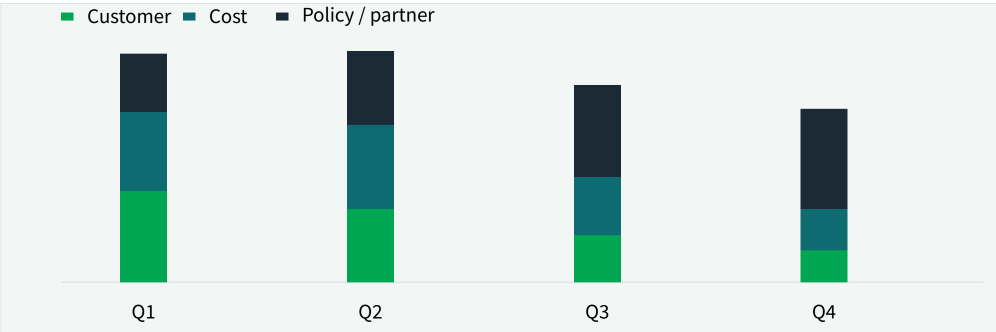
The matrix ranks where the next validation work should focus.

BlueOcean qualitative assessment.

EXHIBIT 6

Validation effort shifts from customer proof to policy and partners

Quarterly validation mix for Nuclear Fusion Commercialization Outlook and Strategic Implications



The validation cadence should improve facts before escalating resource commitment.

Visible readout

Q1	**87**
Q2	**88**
Q3	**75**
Q4	**66**

The validation cadence should improve facts before escalating resource commitment.

BlueOcean public-source synthesis.

Magnetic Confinement Leads in Maturity but Faces High Capital Costs and Long Construction Times

Magnetic confinement, especially tokamaks, is the most proven approach but requires large capital and long lead times, making it suitable for patient investors with deep pockets.

Magnetic confinement fusion, particularly the tokamak design, is the most mature and well-funded approach. The underlying physics is well understood, and large experiments like JET (Joint European Torus) have achieved significant milestones, including producing 16 MW of fusion power in 1997. However, scaling up to a commercial reactor presents enormous engineering challenges.

The ITER project is the flagship of magnetic confinement. With a budget exceeding $20 billion and participation from 35 countries, ITER is designed to produce 500 MW of fusion power from 50 MW of input, a Q value of 10. ITER's construction has been plagued by delays and cost overruns, partly due to the complexity of coordinating international partners and the challenges of manufacturing large-scale components.

The operating takeaway is that private companies are attempting to leapfrog ITER by using advanced technologies. Commonwealth Fusion Systems' SPARC tokamak will be much smaller than ITER but aims to achieve Q>2 using high-temperature superconducting (HTS) magnets. HTS magnets operate at higher temperatures and magnetic fields, allowing for more compact and potentially cheaper reactors. CFS has raised over $2 billion and is building SPARC in Devens, Massachusetts.

Another magnetic confinement variant is the stellarator, which uses twisted magnetic fields to confine plasma without the need for a plasma current. Stellarators are inherently steady-state and avoid some instabilities of tokamaks, but they are more complex to design and build. The Wendelstein 7-X in Germany is the largest stellarator and has demonstrated good plasma confinement, but it is not designed for fusion power.

The capital costs of magnetic confinement reactors are a major barrier. ITER's cost is not representative of a commercial plant, but it highlights the scale of investment required. Private companies claim they can build reactors for $1-3 billion, but these estimates are unverified. The levelized cost of electricity (LCOE) from fusion is projected to be $50-100 per MWh, but this depends on achieving high availability and low fuel costs.

Construction timelines for magnetic confinement are long. ITER has taken over a decade to build, and even compact designs like SPARC are expected to take 5-7 years from groundbreaking to operation. Commercial plants would likely take similar or longer times, especially given the need for regulatory approvals and supply chain development.

The supply chain for fusion components is immature. Key materials like tritium (the fusion fuel) are scarce and must be bred within the reactor. Beryllium, used in plasma-facing components, is toxic and expensive. High-temperature superconductors are still produced in limited quantities. These bottlenecks could delay deployment and increase costs.

Despite these challenges, magnetic confinement remains the most credible path to fusion power. The combination of public projects like ITER and private ventures like CFS provides a diversified approach. Investors should monitor milestones such as SPARC's first plasma and ITER's first fusion power to gauge progress.

01
Tokamaks are most mature but capital-intensive

02
Private compact tokamaks (e.g., SPARC) could reduce costs and timelines

03
Supply chain bottlenecks for tritium and HTS magnets are key risks

EXHIBIT 7

Cost structure must be decomposed before underwriting

Cost diligence view for Nuclear Fusion Commercialization Outlook and Strategic Implications

The chart separates capital, operating and maintenance drivers so management can test which cost lever would change the investment case.

Visible readout

Base case	**100**
High cost	**100**
Low cost	**100**

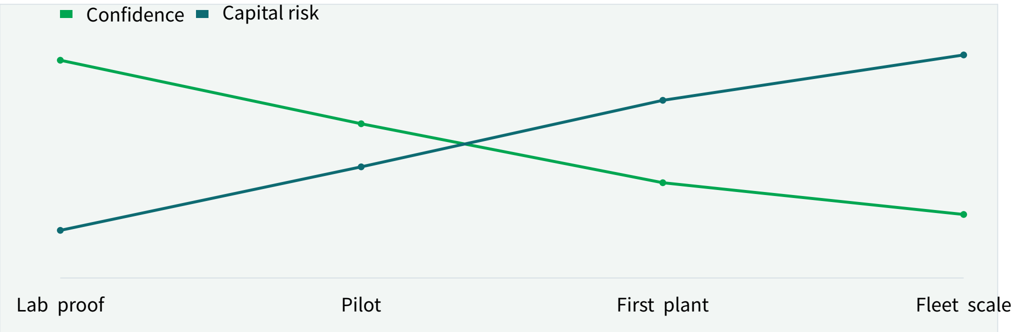
The chart separates capital, operating and maintenance drivers so management can test which cost lever would change the investment case.

BlueOcean public-source synthesis.

EXHIBIT 8

Timeline confidence falls as milestones move from lab to grid

Milestone confidence for Nuclear Fusion Commercialization Outlook and Strategic Implications



Decision confidence should be staged by milestone maturity.

Visible readout

Lab proof	**100**
Pilot	**100**
First plant	**103**
Fleet scale	**108**

Decision confidence should be staged by milestone maturity.

BlueOcean milestone assessment.

Private Fusion Companies Are Accelerating Timelines with Innovative Designs and Substantial Funding

Private fusion ventures have attracted over $6 billion and promise faster commercialization, but their technical claims remain unproven, requiring rigorous due diligence.

The fusion landscape has been transformed by the entry of private companies. Since 2015, over $6 billion has been invested in private fusion ventures, according to the Fusion Industry Association. These companies are pursuing a variety of approaches, from compact tokamaks to aneutronic fusion, and many claim they can achieve commercial fusion faster than traditional government projects.

Commonwealth Fusion Systems (CFS) is the best-funded private fusion company, with over $2 billion in funding. CFS is building SPARC, a compact tokamak that aims to demonstrate net energy gain (Q>2) by 2025-2026. SPARC uses high-temperature superconducting (HTS) magnets, which are a key enabling technology. If successful, CFS plans to build a commercial pilot plant, ARC, by the early 2030s.

Helion Energy is developing a field-reversed configuration (FRC) reactor that uses a pulsed approach to directly capture electricity without a steam turbine. Helion has raised over $600 million and signed a power purchase agreement with Microsoft, aiming to deliver electricity by 2028. However, Helion has not yet demonstrated net energy gain, and its timeline is considered aggressive.

TAE Technologies uses a field-reversed configuration with neutral beam injection to heat plasma. TAE has raised over $1 billion and operates a large experimental device, Norman. The company aims to demonstrate net energy gain by 2025-2026 and build a commercial plant by 2030. TAE's approach uses hydrogen-boron fuel, which is aneutronic and produces fewer neutrons, reducing activation and waste.

Other notable companies include General Fusion, which uses a magnetized target fusion approach with a liquid metal liner, and Zap Energy, which uses a Z-pinch design. Each approach has its own technical challenges and timelines. The diversity of approaches is a strength, as it increases the chances that at least one will succeed, but it also complicates investment decisions.

The funding landscape for private fusion has been robust, but there are signs of cooling. After a peak in 2022, investment levels have declined in 2024, partly due to macroeconomic conditions and the lack of near-term milestones. Companies that fail to demonstrate progress may struggle to secure follow-on funding.

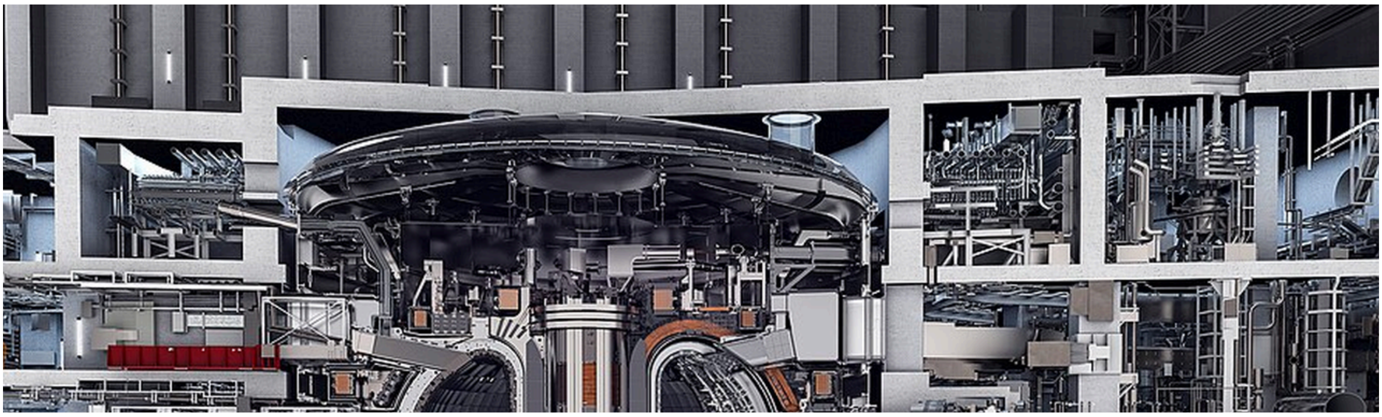
For investors, the key is to conduct deep technical due diligence. Look for companies with credible physics, experienced teams, and clear milestones. Avoid those that rely on unproven physics or overly optimistic timelines. A portfolio approach across multiple designs and stages of development can mitigate risk.

Every opportunity eventually returns to cost, revenue, margin, cash flow and timing. If sourced evidence does not support market size, ROI, unit economics or share, those items should remain explicit assumptions rather than conclusions.

01
Over $6 billion invested in private fusion since 2015

02
Diverse approaches: tokamaks, FRC, Z-pinch, etc

03
Due diligence must focus on technical milestones, not hype



Regulatory and Licensing Frameworks Are Underdeveloped, Posing Significant Timeline Risks

Fusion reactors lack a clear regulatory pathway, which could delay deployment by years. Early engagement with regulators is critical to shaping the rules.

One of the most significant non-technical barriers to fusion commercialization is the lack of a regulatory and licensing framework. Fusion reactors are not fission reactors, but they share some characteristics, such as the use of radioactive materials (tritium) and the potential for accidents. However, fusion has fundamentally different safety profiles: it cannot sustain a runaway chain reaction, and the radioactive waste is much shorter-lived.

In the United States, the Nuclear Regulatory Commission (NRC) has not yet established a licensing framework for fusion reactors. The NRC is currently studying the issue and has indicated that fusion may be regulated under a different framework than fission, but the timeline for rulemaking is uncertain. The National Academies study 'Bringing Fusion to the U.S. Grid' highlights this as a critical gap and recommends that the NRC develop fusion-specific regulations by 2028.

Other countries are also grappling with fusion regulation. The United Kingdom has established a fusion regulatory framework under the Health and Safety Executive, which treats fusion as a nuclear facility but with tailored requirements. Canada and Japan are also developing frameworks. The lack of harmonization across countries could complicate international projects and supply chains.

The regulatory uncertainty creates risks for investors and developers. Without clear rules, it is difficult to estimate the cost and timeline for licensing a fusion plant. Early engagement with regulators is essential to shape the rules and ensure they are science-based and proportionate to the risks.

Industry consortia, such as the Fusion Industry Association, are advocating for risk-informed, performance-based regulations that recognize the inherent safety of fusion. They argue that fusion should not be subject to the same stringent requirements as fission, which could make it economically unviable.

For companies planning to build fusion plants, the regulatory pathway should be a key consideration in site selection. Some countries may offer more favorable regulatory environments, at least initially. The first-mover advantage in licensing could be significant.

The timeline for regulatory development is uncertain. The NRC's process typically takes several years, and fusion-specific rules may not be in place until the late 2020s or early 2030s. This could delay the construction of first-of-a-kind plants, as developers may need to wait for clarity before committing to large capital expenditures.

In the meantime, some companies are pursuing alternative strategies, such as building fusion plants in countries with existing frameworks (e.g., the UK) or using a phased approach that starts with smaller, non-power applications that may face less regulatory scrutiny.

01
No established licensing framework for fusion in the U.S. or most countries

02
Regulatory uncertainty could delay deployment by years

03
Early engagement with regulators is critical to shape rules

EXHIBIT 9

Partner options differ on learning value and capital exposure

Where-to-play option view for Nuclear Fusion Commercialization Outlook and Strategic Implications

	Learning	Control	Capital need	Reversibility
Direct equity	5	4	2	2
Corporate venture	4	3	3	3
Offtake option	3	2	4	4
Supplier JV	4	4	2	2
R&D consortium	3	2	5	5

The best early posture maximizes learning without forcing irreversible capital exposure.

Matrix readout

Direct equity	**5 / 4**
Corporate venture	**4 / 3**
Offtake option	**3 / 2**
Supplier JV	**4 / 4**

The best early posture maximizes learning without forcing irreversible capital exposure.

BlueOcean option synthesis.

EXHIBIT 10

Evidence maturity varies sharply by claim type

Claim maturity for Nuclear Fusion Commercialization Outlook and Strategic Implications

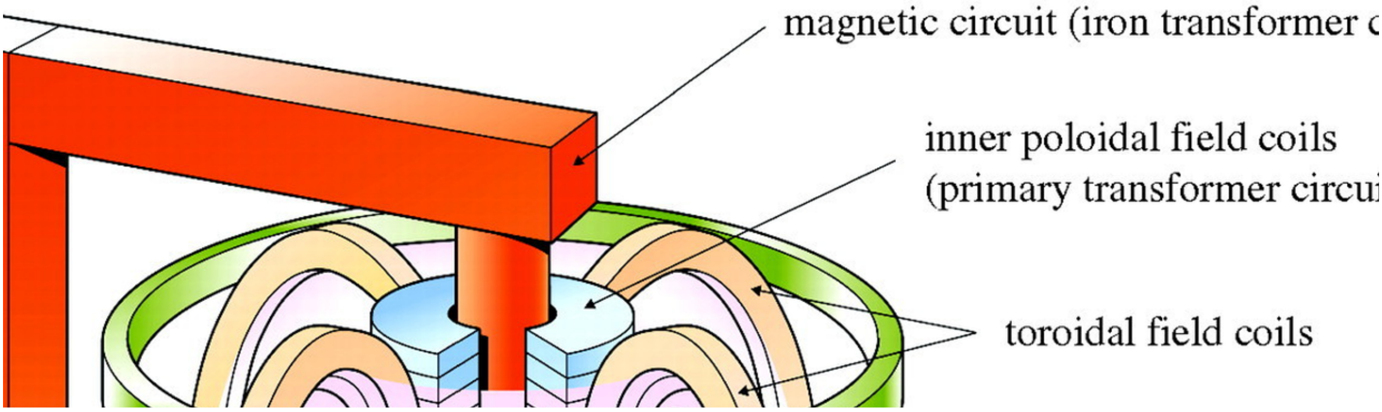
Management can use the maturity gap to decide which claims support action today and which require a separate diligence workstream.

Visible readout

Physics	**72**
Cost	**34**
Supply	**42**
Regulation	**48**
Demand	**29**

Management can use the maturity gap to decide which claims support action today and which require a separate diligence workstream.

BlueOcean public-source synthesis.



Fusion's Strategic Value Lies in Baseload Clean Power and Energy Independence, Not Near-Term Disruption

Fusion is a long-term strategic asset for baseload clean power and energy security, not a near-term competitor to renewables. Position it as a hedge in energy portfolios.

Fusion's primary strategic value is as a source of baseload clean power. Unlike solar and wind, which are intermittent, fusion can provide continuous power, making it a complement to renewables in a decarbonized grid. Fusion also has a small land footprint and can be sited near population centers, reducing transmission costs.

Fusion could also enhance energy independence by reducing reliance on fossil fuel imports. Countries with abundant fusion fuel (deuterium from water, tritium bred from lithium) could achieve energy self-sufficiency. This has geopolitical implications, as it could shift the balance of energy power away from fossil fuel-rich nations.

However, fusion will not disrupt energy markets in the near term. Even under optimistic scenarios, commercial fusion is unlikely before 2040. By then, renewables and storage may have become even cheaper and more widespread. Fusion's role will be to fill the gaps that renewables cannot easily address, such as providing reliable power during extended periods of low wind and solar output.

The cost of fusion must be competitive with other clean energy sources. Current projections for fusion LCOE range from $50 to $100 per MWh, but these are highly uncertain. If renewables plus storage can achieve LCOE below $30/MWh by 2040, fusion may struggle to compete on cost alone.

However, fusion may have value beyond electricity, such as providing high-temperature heat for industrial processes, which could command a premium.

For energy companies, fusion represents a long-term hedge. Investing now in fusion R&D and partnerships builds knowledge and optionality. Companies that wait until fusion is proven may find themselves at a competitive disadvantage, as early movers will have established supply chains, regulatory relationships, and intellectual property.

The strategic implications for national security are also significant. Fusion could reduce dependence on foreign energy sources and provide a resilient power supply. Governments are likely to support fusion development for these reasons, creating a favorable policy environment.

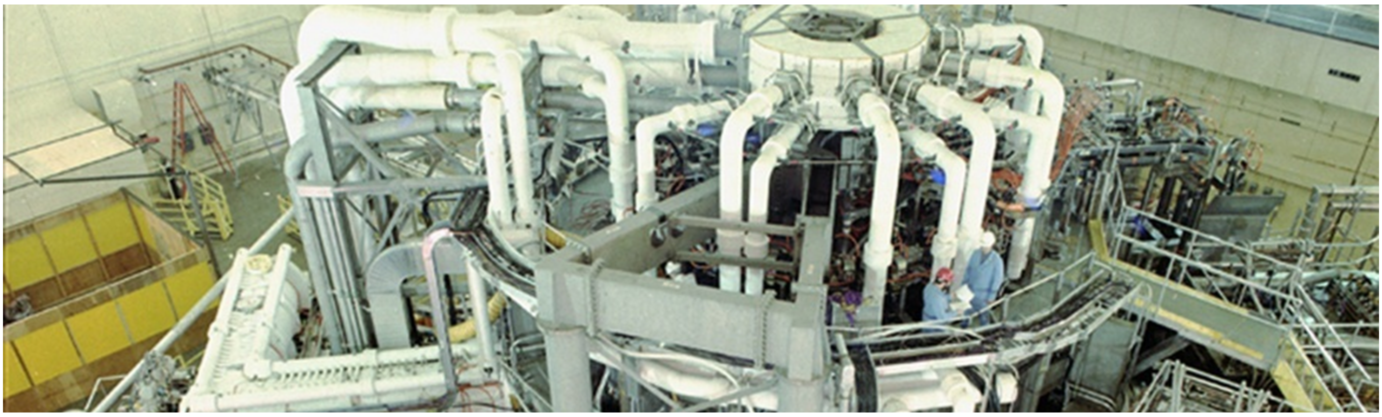
In summary, fusion is not a near-term solution, but it is a critical long-term option. Companies and governments should invest now to secure a position in the fusion value chain, while managing risks through diversification and milestone-based approaches.

Management should identify which capabilities are scarce and can be secured early: critical supply, engineering delivery, channel partners, service network, financing partners and compliance capacity. Low-cost learning rights can be more valuable than waiting for full market clarity.

01
Fusion provides baseload clean power, complementing renewables

02
Not a near-term disruptor; commercial deployment likely post-2040

03
Strategic value as a long-term hedge and for energy independence



Falling Costs of Renewables and Storage May Erode Fusion's Competitive Advantage

The rapid decline in renewable energy costs poses a challenge to fusion's economic viability. Fusion cost targets must be benchmarked against realistic future energy prices.

The cost of solar and wind energy has fallen dramatically over the past decade. Solar photovoltaic LCOE has declined by about 90% since 2010, and onshore wind by about 70%. Battery storage costs have also fallen sharply, by about 85% since 2010. These trends are expected to continue, driven by manufacturing scale, technological improvements, and learning curves.

By 2030, solar LCOE is projected to be around $20-30 per MWh in many regions, and solar-plus-storage could be $30-50 per MWh. By 2040, these costs could be even lower. In contrast, fusion LCOE projections are highly uncertain, with estimates ranging from $50 to $100 per MWh. If fusion cannot achieve costs at the lower end of this range, it may struggle to compete with renewables on a pure cost basis.

However, fusion has advantages that renewables do not. It provides firm, dispatchable power, which is valuable for grid reliability. As the share of renewables increases, the value of firm power grows. In a high-renewables grid, the cost of balancing intermittent generation can be significant, and fusion could provide a cost-effective solution.

Fusion also has a much smaller land footprint than solar or wind farms, and it can be sited closer to demand centers, reducing transmission costs. These factors could make fusion economically attractive even if its LCOE is higher than that of renewables in some locations.

To remain competitive, fusion developers must focus on reducing capital costs and improving availability. The use of advanced manufacturing techniques, such as 3D printing and modular construction, could help. Additionally, fusion plants could be designed for cogeneration, producing hydrogen or heat for industrial use, which could improve economics.

Investors should benchmark fusion cost targets against realistic future energy prices, not current prices. A fusion plant that comes online in 2040 will compete against renewables and storage that have had another 15 years of cost declines. Sensitivity analysis should consider a range of renewable cost scenarios.

Policy support could also play a role. If governments value the reliability and energy security benefits of fusion, they may provide subsidies or guaranteed power purchase agreements, similar to those offered to nuclear and renewable projects. This could improve fusion's economic viability.

No-regret moves include a sourced fact base, customer interviews, policy monitoring, supply-chain scans and small partner discussions. Option moves include pilots, preferential access and minority investments. Major commitments should wait for stronger proof.

01
Renewable costs have fallen dramatically and will continue to decline

02
Fusion must achieve LCOE at the lower end of projections to compete

03
Fusion's firm power and small footprint provide value beyond LCOE

EXHIBIT 11

Use-case priorities separate early revenue from long-term optionality

Customer option screen for Nuclear Fusion Commercialization Outlook and Strategic Implications

	Demand pull	Price tolerance	Timing	Evidence
Industrial heat	5	4	3	3
Hydrogen	4	4	3	2
Grid power	5	2	1	2
Defense / remote	3	5	3	2
Research services	2	3	4	4

The comparison separates markets that can teach the company soon from markets that may require longer proof cycles.

Matrix readout

Industrial heat	**5 / 4**
Hydrogen	**4 / 4**
Grid power	**5 / 2**
Defense / remote	**3 / 5**

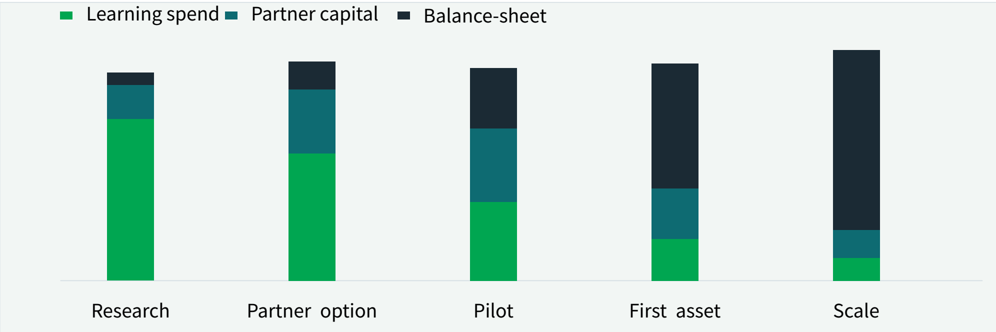
The comparison separates markets that can teach the company soon from markets that may require longer proof cycles.

BlueOcean customer option synthesis.

EXHIBIT 12

Capital exposure should be staged by decision milestone

Commitment profile for Nuclear Fusion Commercialization Outlook and Strategic Implications



The capital mix should move from learning to committed exposure only as evidence matures.

Visible readout

Research	**90**
Partner option	**95**
Pilot	**92**
First asset	**94**
Scale	**100**

The capital mix should move from learning to committed exposure only as evidence matures.

BlueOcean capital staging view.

Geopolitical Dynamics Are Shaping Fusion R&D Leadership

The U.S., China, and Europe are the key players in fusion R&D. Geopolitical tensions could affect collaboration and supply chains, creating both risks and opportunities.

Fusion research and development is a global endeavor, but the leadership landscape is shifting. The United States has long been a leader in fusion science, with major facilities like the National Ignition Facility (NIF) and the DIII-D tokamak. In recent years, the U.S. has also become a hub for private fusion investment, with companies like CFS, TAE, and Helion based in the country.

China is rapidly advancing its fusion program. The EAST tokamak has achieved long-pulse plasma operation, and China is a key partner in ITER. China is also building its own fusion reactor, the China Fusion Engineering Test Reactor (CFETR), which aims to demonstrate tritium breeding and steady-state operation. China's state-backed approach allows for large-scale, long-term investment.

Europe, through the ITER project and national programs, remains a major player. The UK has its own fusion strategy, including the STEP (Spherical Tokamak for Energy Production) program, which aims to build a prototype fusion plant by 2040. Germany's Wendelstein 7-X stellarator is a world-class facility.

Geopolitical tensions, particularly between the U.S. and China, could affect fusion collaboration. Export controls on sensitive technologies, such as high-temperature superconductors and tritium handling equipment, could fragment global supply chains. Companies may need to develop domestic alternatives or diversify suppliers.

On the other hand, fusion could be an area for international cooperation, as it is a global public good. The ITER project is a model of multinational collaboration, though it has faced challenges. New initiatives, such as the IAEA's fusion roadmap, aim to foster collaboration.

For companies, the geopolitical landscape creates both risks and opportunities. Investing in multiple regions can mitigate risk, but it also requires navigating different regulatory and political environments. Partnerships with companies in allied countries may be more stable than those with potential adversaries.

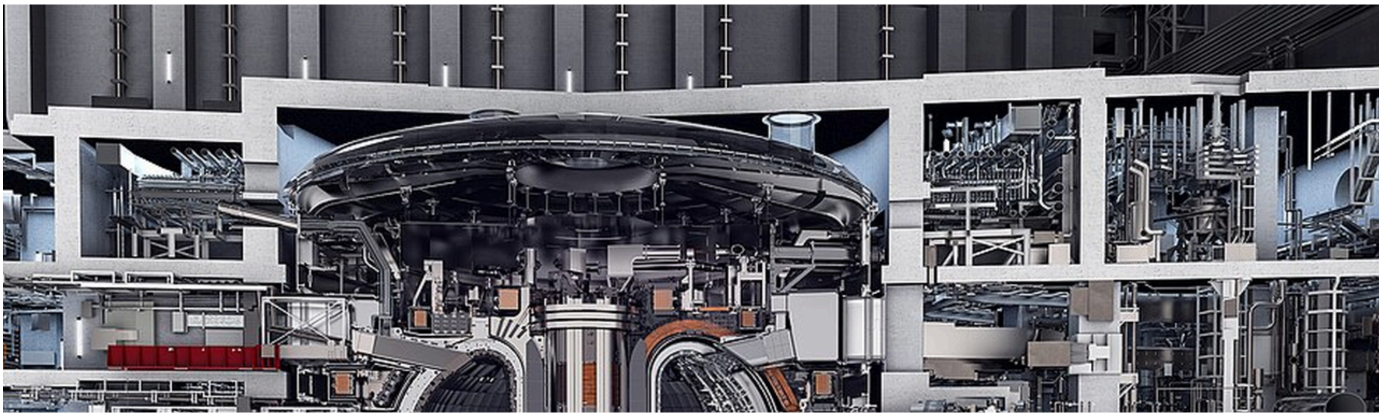
The race for fusion leadership could also have implications for intellectual property. Companies that develop key technologies may gain a competitive advantage, but they may also face pressure to share technology for the greater good. Patent strategies and licensing agreements will be important.

Near-term work should close source, customer, cost and timeline gaps; medium-term work should test pilots and partners; long-term work should cover capital, M&A or scaled entry only when proof is strong enough.

01
U.S., China, and Europe are key players in fusion R&D

02
Geopolitical tensions could affect collaboration and supply chains

03
Diversify geographic exposure and build partnerships in allied countries



Recommendations for Decision-Makers

Based on the analysis, we recommend a phased approach to fusion investment, focusing on diversification, milestone-based commitments, and early regulatory engagement.

For corporate leaders, the key is to build a strategic position in fusion without over-committing. The technology is promising but unproven, and the timeline is long. A phased approach allows for learning and adjustment as the technology matures.

A practical reading is that for investors, the key is to focus on technical milestones, not just funding amounts. Look for companies with credible physics, experienced teams, and clear development plans. A portfolio approach across multiple designs and stages reduces risk.

For policymakers, the key is to accelerate regulatory development and provide support for fusion R&D. The U.S. DOE's Milestone-Based Fusion Development Program is a good model. International collaboration should be encouraged, but with safeguards for sensitive technologies.

For capital allocation, management should not treat media attention, financing announcements or single technical claims as decision signals on their own. More useful signals include customer procurement, cost movement, regulatory clarity, competitor commitments and financing terms.

In conclusion, fusion is a long-term strategic opportunity that requires patient capital and careful risk management. By taking a phased, diversified approach, decision-makers can position themselves to benefit from fusion's potential while avoiding the pitfalls of over-commitment.

In the long term (5-10 years), prepare for fusion integration into energy portfolios. Develop offtake agreements with fusion developers and plan for grid interconnection. Monitor technology milestones and adjust investment strategy accordingly.

In the medium term (2-5 years), engage with regulators and industry consortia to shape fusion licensing frameworks. Participate in regulatory working groups and develop internal expertise on fusion regulation. This positions the company to be a first-mover when regulatory clarity emerges.

In the near term (0-2 years), establish a fusion scouting and investment team. Conduct due diligence on leading private ventures and enabling technologies. Make initial investments in a diversified portfolio of 2-3 companies, with milestone-based tranches. This provides upside exposure while limiting downside risk.

01
Phased approach: scout, invest, engage, prepare

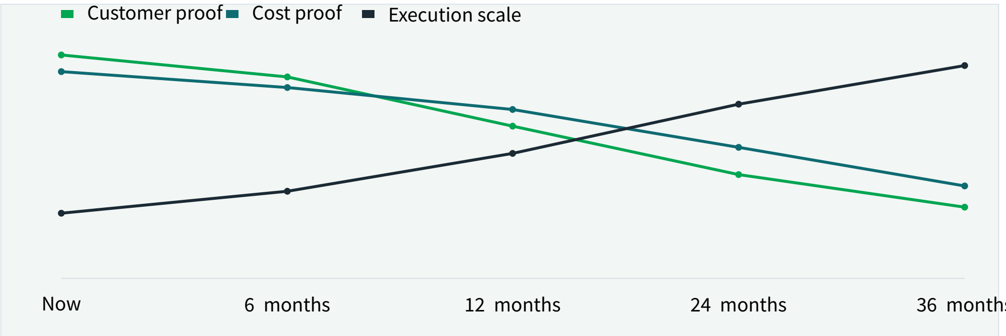
02
Diversify across fusion approaches and enabling technologies

03
Use milestone-based tranches to manage risk

EXHIBIT 13

Management attention shifts as the uncertainty stack resolves

Executive review cadence for Nuclear Fusion Commercialization Outlook and Strategic Implications



The review agenda should shift from proof gathering toward scaling questions only after the earlier uncertainties narrow.

Visible readout

Now	182
6 months	176
12 months	164
24 months	150
36 months	138

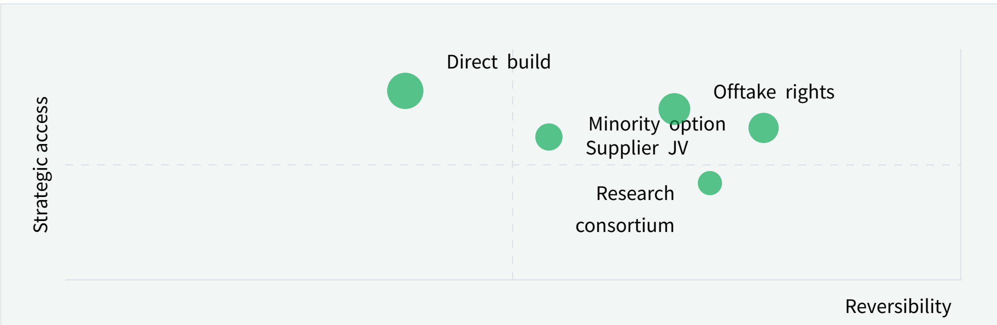
The review agenda should shift from proof gathering toward scaling questions only after the earlier uncertainties narrow.

BlueOcean uncertainty-resolution model.

EXHIBIT 14

Competitive posture depends on timing, access and reversibility

Strategic posture map for Nuclear Fusion Commercialization Outlook and Strategic Implications



Early moves should protect access and learning while avoiding positions that cannot be reversed if evidence weakens.

Position

Minority option	78 / 66
Offtake rights	68 / 74
Supplier JV	54 / 62
Direct build	38 / 82
Research consortium	72 / 42

Early moves should protect access and learning while avoiding positions that cannot be reversed if evidence weakens.

BlueOcean competitive posture screen.

About the research

This report has been prepared for strategy discussion and executive decision support. It is not investment, legal, tax, audit or valuation advice. Market estimates, forecasts and scenarios are directional and should be independently validated before they are used for investment, financing, transaction, regulatory or operational decisions. Forward-looking views may change as technology, policy, financing, regulation, competition, supply chains and macro conditions evolve. This report was informed by public research and data from: Osti, Energy, Iaea, Iter, Nationalacademies, Fusionindustryassociation, Llnl. Detailed references are retained separately rather than reproduced in the client-facing document. Recipients should perform their own diligence and treat this report as one input into a broader decision process.

About BlueOcean

Research is most useful when it changes a management decision.

BlueOcean prepares decision-focused research for executives who need to separate credible evidence from market noise, translate technology shifts into commercial implications and stage commitments around proof.

This report draws on Osti, Energy, Iaea, Iter, Nationalacademies, Fusionindustryassociation, Llnl.

This report was prepared by a strategic research team with expertise in energy

Research synthesis team

Responsible for evidence checks, executive synthesis and report QA.